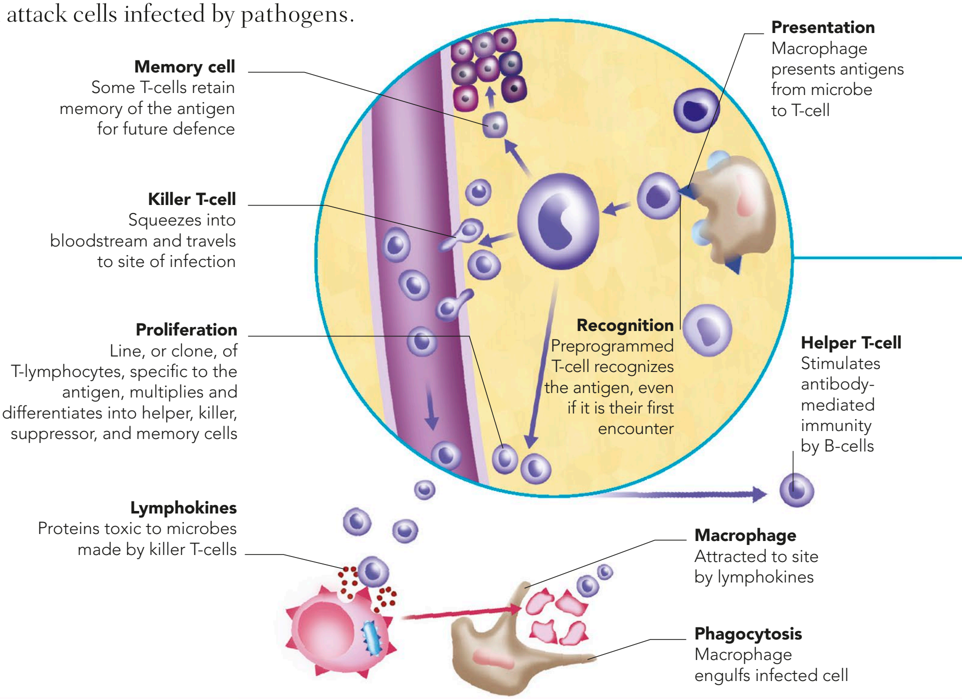
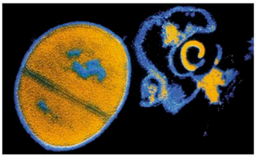
SPECIFIC RESPONSE

The two main types of specific defence – cell-mediated and antibody-mediated immunity – may accompany non-specific reactions such as inflammation, or follow if infection persists. Both depend on the actions of B- and T- lymphocytes. B-cells make protein antibodies known as gammaglobulins, which react against antigens (foreign proteins). Types of T-cells multiply and attack cells infected by pathogens.

CELL-MEDIATED IMMUNITY

Once a T-cell recognizes an antigen, it multiplies rapidly and its offspring form several types. Killer T-cells attack and destroy infected cells, which have the antigen on their surface. Helper T-cells activate both B-cells to help antibody-mediated immunity and macrophages to engulf debris. Suppressor (regulatory) T-cells dampen down the body's immune response after the infection has been dealt with.





LYSED BACTERIUM
Complement dissolves, or lyses, invaders such as bacteria by disrupting their outer membranes (cell shown on right).

COMPLEMENT SYSTEM

More than 25 proteins and related substances in the blood form the complement system, which joins the fight against invading microbes. Once a complement reaction begins, it carries on in a "cascade", with one protein activating the next, and so on. The complement system has several roles. It generally helps to destroy microbes, and prevent them attacking body cells, encourages the activity of white cells, widens blood vessels, and clears away the antigen–antibody complexes.